KAKSHA PRACTICE PAPER · CLASS 9

Science Mock Test 1

Time allowed: 3 hours **Maximum marks: 80**

Five sections A–E. Section A: 20 one-mark questions (16 MCQs and 4 assertion–reason). Section B: six 2–mark very short answer questions. Section C: seven 3–mark short answer questions. Section D: three 5–mark long answer questions. Section E: three case-based questions of 4 marks.

Section A

Multiple-choice questions (Q1–16) and assertion–reason questions (Q17–20). Choose the correct option.

1. The rate of diffusion is highest in **[1]**

(a) solids (b) liquids
(c) gases (d) all of them equally

2. The boiling point of water at normal atmospheric pressure, on the Kelvin scale, is **[1]**

(a) 100 K (b) 273 K
(c) 373 K (d) 473 K

3. Which of the following will show the Tyndall effect? **[1]**

(a) Salt solution (b) Sugar solution
(c) Copper sulphate solution (d) Milk

4. Which of the following is a chemical change? **[1]**

(a) Melting of ice (b) Dissolving sugar in water
(c) Rusting of iron (d) Boiling of water

5. The molar mass of carbon dioxide (C = 12 u, O = 16 u) is **[1]**

(a) 28 g/mol (b) 44 g/mol
(c) 32 g/mol (d) 60 g/mol

6. The electronic configuration of sodium (atomic number 11) is **[1]**

(a) 2, 9 (b) 2, 8
(c) 8, 2, 1 (d) 2, 8, 1

7. Isotopes of an element have the same **[1]**

(a) atomic number (b) mass number
(c) number of neutrons (d) atomic mass

8. Which organelle is present in plant cells but absent in animal cells? **[1]**

(a) Mitochondria
(b) Ribosomes
(c) Plastids
(d) Nucleus

9. The plant tissue that provides flexibility to leaves and stems, allowing them to bend without breaking, is **[1]**

(a) parenchyma
(b) collenchyma
(c) sclerenchyma
(d) xylem

10. Blood is a type of **[1]**

(a) epithelial tissue
(b) muscular tissue
(c) nervous tissue
(d) connective tissue

11. The area under a velocity–time graph gives the **[1]**

(a) acceleration
(b) speed
(c) displacement
(d) force

12. The SI unit of momentum is **[1]**

(a) kg m/s^2
(b) kg m/s
(c) N/s
(d) J s

13. The value of the universal gravitational constant G is **[1]**

(a) 6.673×10^{-11} N m^2 kg^{-2}
(b) 9.8 N m^2 kg^{-2}
(c) 6.673×10^{-11} m/s^2
(d) 9.8×10^{-11} N m^2 kg^{-2}

14. One kilowatt hour (1 kWh) is equal to **[1]**

(a) 1000 J
(b) 3600 J
(c) 3.6×10^3 J
(d) 3.6×10^6 J

15. The audible range of hearing for humans is **[1]**

(a) 2 Hz to 2000 Hz
(b) 20 Hz to 20,000 Hz
(c) 200 Hz to 200,000 Hz
(d) above 20,000 Hz

16. Which of the following is an exotic (foreign) breed of cattle? **[1]**

(a) Red Sindhi
(b) Sahiwal
(c) Jersey
(d) Gir

17. Assertion (A): A ball thrown vertically upwards has zero velocity at its highest point.
Reason (R): The acceleration of the ball at its highest point is zero. **[1]**

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

18. Assertion (A): An object weighs less on the Moon than on the Earth. Reason (R): The acceleration due to gravity on the Moon is about one-sixth of that on the Earth. **[1]**

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

19. Assertion (A): Sound travels faster in air than in steel. Reason (R): Sound needs a material medium to travel. **[1]**

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

20. Assertion (A): The cell is the structural and functional unit of life. Reason (R): Robert Hooke observed cells in a thin slice of cork in 1665. **[1]**

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Section B

Very short answer questions.

21. Why do we see water droplets on the outer surface of a glass containing ice-cold water? **[2]**

22. Calculate the number of moles and the number of atoms in 52 g of helium. (Atomic mass of He = 4 u, $N_a = 6.022 \times 10^{23}$) **[2]**

23. Classify the following into homogeneous and heterogeneous mixtures: soda water, wood, air, soil, vinegar, filtered tea. **[2]**

24. Why is the plasma membrane called a selectively permeable membrane? Name the process by which water moves across it. **[2]**

25. A car accelerates uniformly from 18 km/h to 36 km/h in 5 s. Find its acceleration and the distance covered in this time. **[2]**

26. An object of mass 10 kg is at a height of 6 m above the ground. Find its potential energy. What will be its kinetic energy when it has fallen halfway down? ($g = 9.8\ m/s^2$) **[2]**

Section C

Short answer questions.

27. Give reasons: (i) A solid has a definite shape. (ii) Gases can be compressed easily. (iii) Naphthalene balls disappear with time without leaving any residue. **[3]**

28. Write the chemical formulae of (i) aluminium sulphate (ii) calcium hydroxide (iii) ammonium chloride, showing the valencies used. **[3]**

29. State any three postulates of Bohr's model of the atom. **[3]**

30. Differentiate between striated, unstriated and cardiac muscles on the basis of shape of cells, nuclei and location. **[3]**

31. A bullet of mass 20 g is fired horizontally with a velocity of 150 m/s from a pistol of mass 2 kg. Find the recoil velocity of the pistol. Which law is used? **[3]**

32. State Archimedes' principle. Give two of its applications. **[3]**

33. What is an echo? A person claps near a cliff and hears the echo after 2 s. If the speed of sound is 346 m/s, how far is the cliff? **[3]**

Section D

Long answer questions.

34. Name the technique you would use to separate each of the following, giving one reason in each case: (i) ammonium chloride from common salt (ii) cream from milk (iii) different coloured dyes in black ink (iv) kerosene from water (v) acetone from water. **[5]**

35. Using a velocity–time graph, derive (i) $v = u + at$ and (ii) $s = ut + \tfrac{1}{2}at^2$ for an object moving with uniform acceleration. **[5]**

36. (a) Differentiate between mixed cropping and intercropping. (b) Describe three methods of controlling weeds, insect pests and diseases in crops. (c) Name two biotic and two abiotic factors responsible for losses of stored grain. **[5]**

Section E

Case-study questions (1 + 1 + 2 marks).

37. Case study — Two kinds of chlorine. Chlorine found in nature is a mixture of two atoms, $^{35}_{17}\mathrm{Cl}$ and $^{37}_{17}\mathrm{Cl}$, in the ratio 3 : 1. Such atoms are used widely; for example, an isotope of cobalt is used to treat cancer. (i) What are isotopes? (1) (ii) Calculate the average atomic mass of chlorine. (1) (iii) Write the number of protons and neutrons in each chlorine atom, and give one more use of isotopes. (2) **[4]**

38. Case study — The swelling raisins. Riya soaked dry raisins in plain water. After a few hours they swelled up. When she put the swollen raisins in a concentrated sugar solution, they shrank. Her teacher said this is due to the movement of water across the cell membrane. (i) Name the process of movement of water across a selectively permeable membrane. (1) (ii) What happens to a plant cell placed in a hypertonic solution? (1) (iii) Give two differences between a plant cell and an animal cell. (2) **[4]**

39. Case study — Pressure and floating. School bags have broad straps. Camels walk easily on sand. An iron nail sinks in water, but a cork floats, and a huge ship made of iron also floats. (i) Define pressure and give its SI unit. (1) (ii) Why do school bags have broad straps? (1) (iii) Why does an iron nail sink while a cork floats in water? (2) **[4]**

Answer key and marking points

1. (c) gases

2. (c) 373 K

3. (d) Milk

4. (c) Rusting of iron

5. (b) 44 g/mol

6. (d) 2, 8, 1

7. (a) atomic number

8. (c) Plastids

9. (b) collenchyma

10. (d) connective tissue

11. (c) displacement

12. (b) kg m/s

13. (a) 6.673×10^{-11} N m^2 kg^{-2}

14. (d) 3.6×10^6 J

15. (b) 20 Hz to 20,000 Hz

16. (c) Jersey

17. (c) A is true but R is false

18. (a) Both A and R are true, and R is the correct explanation of A

19. (d) A is false but R is true

20. (b) Both A and R are true, but R is not the correct explanation of A

21. Water vapour present in the air comes in contact with the cold outer surface of the glass. (1)
It loses energy, cools and condenses into liquid water droplets. (1)

22. Number of moles = given mass / molar mass = 52/4 = 13 mol. (1)
Number of atoms = $13 \times 6.022 \times 10^{23} = 7.83 \times 10^{24}$ atoms. (1)

23. Homogeneous: soda water, air, vinegar, filtered tea. (1)
Heterogeneous: wood, soil. (1)

24. It allows the entry and exit of only some materials (such as water, O_2, CO_2) and prevents the movement of others. (1)
Water moves across it by osmosis (from a region of higher water concentration to lower water concentration). (1)

25. u = 18 km/h = 5 m/s, v = 36 km/h = 10 m/s. a = (v – u)/t = 5/5 = 1 m/s^2. (1)
$s = ut + \frac{1}{2}at^2 = 5 \times 5 + \frac{1}{2} \times 1 \times 25 = 25 + 12.5 = 37.5$ m. (1)

26. PE = mgh = $10 \times 9.8 \times 6 = 588$ J. (1)
Halfway down, PE = 294 J; by conservation of energy, KE = 588 – 294 = 294 J. (1)

27. (i) Particles of a solid are closely packed with strong forces of attraction and very little space, so they cannot move freely. (1)
(ii) Gas particles have large spaces between them, so they can be pushed closer together. (1)
(iii) Naphthalene undergoes sublimation – it changes directly from solid to vapour. (1)

28. (i) Al^{3+} and SO_4^{2-}: $Al_2(SO_4)_3$. (1)
(ii) Ca^{2+} and OH^-: $Ca(OH)_2$. (1)
(iii) NH_4^+ and Cl^-: NH_4Cl. (1)

29. Only certain special orbits, known as discrete orbits (energy levels or shells), are allowed inside the atom. (1)
While revolving in these discrete orbits, the electrons do not radiate energy. (1)
The orbits are represented by letters K, L, M, N… or numbers n = 1, 2, 3, 4… (1)

30. Striated (skeletal/voluntary): long, cylindrical, unbranched cells with light and dark bands; multinucleate; attached to bones. (1)
Unstriated (smooth/involuntary): spindle-shaped cells without striations; uninucleate; found in walls of the alimentary canal, iris, ureters, blood vessels. (1)
Cardiac: cylindrical, branched cells with faint striations; uninucleate; found only in the heart, show rhythmic contraction throughout life. (1)

31. By conservation of momentum, total momentum before firing = 0 = $m_1v_1 + m_2v_2$. (1)
$0.02 \times 150 + 2 \times v = 0 \Rightarrow v = -3/2 = -1.5$ m/s. (1)
The pistol recoils with 1.5 m/s opposite to the bullet. The law of conservation of momentum (a consequence of Newton's third law) is used. (1)

32. When a body is immersed fully or partially in a fluid, it experiences an upward force equal to the weight of the fluid displaced by it. (1)
Applications (any two): designing ships and submarines; lactometers to test purity of milk; hydrometers to find the density of liquids. (2)

33. An echo is the repetition of sound heard due to reflection of sound from a large obstacle (the sensation of sound persists for 0.1 s, so the obstacle must be at least 17.2 m away). (1)
In 2 s, sound travels to the cliff and back: total distance = $346 \times 2 = 692$ m. (1)
Distance of cliff = 692/2 = 346 m. (1)

34. (i) Sublimation – ammonium chloride sublimes on heating, salt does not. (1)
(ii) Centrifugation – spinning makes the lighter cream particles collect at the top. (1)
(iii) Chromatography – dyes dissolve in the solvent to different extents and rise to different heights on the paper. (1)
(iv) Separating funnel – the two immiscible liquids form separate layers; the lower layer (water) is drained off. (1)
(v) Distillation – the miscible liquids have boiling points differing by more than 25 K; acetone (b.p. 56 °C) boils off first. (1)

35. Let the object have initial velocity u (point A on the velocity axis, OA = u) and velocity v (point B) after time t (OC = t). Draw BC ⊥ time axis and AD ∥ time axis, so BC = v, DC = OA = u, AD = OC = t. (1)
(i) Acceleration a = change in velocity / time = BD/AD = (BC – DC)/t = (v – u)/t. (1)
So $v = u + at$. (½)
(ii) Distance s = area under the graph OABC = area of rectangle OADC + area of triangle ABD. (1)
$= u \times t + \frac{1}{2} \times t \times (v - u) = ut + \frac{1}{2} \times t \times at$. (1)
So $s = ut + \frac{1}{2}at^2$. (½)

36. (a) Mixed cropping: two or more crops grown simultaneously on the same land, seeds mixed together (e.g. wheat + gram). Intercropping: two or more crops grown simultaneously in a definite row pattern (e.g. soyabean + maize), each crop can get nutrients separately and products can be

harvested separately. (2)
(b) Any three: mechanical removal of weeds; spraying pesticides (weedicides, insecticides, fungicides) in correct amounts; using resistant varieties; proper seed-bed preparation and timely sowing; crop rotation and summer ploughing. (1½)
(c) Biotic: insects, rodents, fungi, mites, bacteria (any two). Abiotic: inappropriate moisture and temperature (any two). (1½)

37. (i) Atoms of the same element having the same atomic number but different mass numbers. (1)
(ii) Average atomic mass = 35 × 3/4 + 37 × 1/4 = 26.25 + 9.25 = 35.5 u. (1)
(iii) ^{35}Cl: 17 protons, 18 neutrons; ^{37}Cl: 17 protons, 20 neutrons. (1) Use (any one): uranium isotope as fuel in nuclear reactors; iodine isotope in treatment of goitre. (1)

38. (i) Osmosis. (1)
(ii) The cell loses water by osmosis and the cell contents shrink away from the cell wall; this is called plasmolysis. (1)
(iii) Any two: plant cells have a cell wall, animal cells do not; plant cells have plastids (chloroplasts), animal cells do not; plant cells have a large central vacuole, animal cells have small vacuoles or none; animal cells have centrosomes, most plant cells do not. (2)

39. (i) Pressure is thrust (force acting perpendicularly) per unit area; SI unit pascal (Pa) or N/m^2. (1)
(ii) Broad straps increase the area of contact with the shoulder, so the pressure on the shoulders is less and the bag is comfortable to carry. (1)
(iii) The density of iron is greater than that of water, so the weight of the nail is more than the buoyant force (weight of water displaced). (1)
The density of cork is less than that of water, so the upthrust on it is greater than its weight and it floats. (1)